E-Commerce Sales & Customer Analytics

Full Data Analyst Portfolio Project Report

Project Type	End-to-End Data Analytics Project
Scope	Sales • Customers • Products • Returns
Data Scale	8,000 Orders • 500 Customers
Tools	Python • Pandas • MySQL • Excel • Power BI
Dashboard	3-Page Interactive Power BI Report

This report documents the complete workflow used to prepare, analyze, visualize, and interpret an e-commerce dataset for business decision-making.

1. Project Overview

TheE-CommerceSales & Customer Analytics project is an end-to-end data analytics project focused on sales performance, profitability, customer behavior, product performance, and returns.

The workflow combines Python/Pandas for data preparation, MySQL for business analysis, Excel for data handling, and Power BI for interactive reporting.
The final Power BI report contains three analytical pages and dynamic DAX measures. Year and Category slicers were testedto confirmthat thekey KPI cards respond to filters.

2. Business Problem

An e-commerce business needs aclearand interactive view of what drives revenue and profit, which customers and products are most valuable, and where returns are occurring. Raw transaction data makes these questions difficult to answer quickly.

- Identify revenue and profit leaders by category and product.
- Measure customer and order performance.
- Track sales and return trends over time.
- Understand the main reasons for product returns.
- Createinteractive KPIs formanagement reporting.

3. Project Objectives

- Cleanand validatetheraw e-commerce dataset.
- Perform feature engineering for business analysis.
- Use SQL to answer key business questions.
- Build an interactive three-page Power BI dashboard.
- Create dynamic DAX measures for orders and returns.
- Translate findingsinto business insightsand recommendations.

4. Dataset Description

Theproject uses e-commercesales/orderdata together with a returns dataset. The sales data includes order, customer, product, quantity, date, cost, revenue and profit-related fields. The returns data contains returned OrderIDs, return datesand return reasons.The analysis covers 2024 and 2025.

5. Tools & Technologies

- **Python:** Datacleaningandfeature engineering.
- **Pandas:** DataFrame manipulation and transformations.
- **MySQL:** Structured business analysis using SQL.
- **Excel:** Data preparation and validation.
- **Power BI:** Interactive visualization and DAX measures.

Area	Examples of Fields
Orders	Order_ID, Customer_ID, Order_Date, Product_ID, Quantity
Financial	Cost, Revenue, Profit
Customer	Customer_ID, Customer_Name
Product	Product_ID, Product_Name, Category
Time	Order Date, Month, Month Year, Year
Returns	Order_ID, Return_Date, Return_Reason

6. Data Cleaning & Feature Engineering

- Checked and handled missingvalues.
- Checked for duplicate records.
- Standardized data types and date fields.
- Validated numerical fields such as cost, revenue, profit and quantity.
- Created Year, Month and Month Year fields for time analysis.
- Prepared revenue,cost, profitand return-related fields for business reporting.

7. MySQL Business Analysis

- Revenue,cost and profit analysis.
- Category-level sales and profit comparisons.
- Top products by revenue and profit.
- Top customers by revenue and orders.
- Monthly and yearly sales trends.
- Returned-orderanalysis andreturn-ratecalculations.

8. Power BI Dashboard

ThePowerBIreportwasdevelopedinPower BI Web using the semantic model **Ecommerce Project**.

Page 1 — E-Commerce Executive Overview

High-level revenue, profit, orders, return rate, monthly trends and category performance.

Page 2 — Product & Customer Performance

Customer and product performance including total customers, total products, units sold, top customers and top products.

Page 3 — Returns & Profitability Analysis

Total orders, returned orders, return rate, return reasons, returned orders by category, monthly returns and revenue versus cost.

9. DAX Measures

Measure	Definition
Returned Orders	DISTINCTCOUNT(returns[Order_ID])
Total Orders	DISTINCTCOUNT(Sheet1[Order_ID])
Return Rate	DIVIDE([Returned Orders], [Total Orders], 0)

The measures were tested with Year and Category slicers and the KPI cards update dynamically.

10. Key Performance Indicators

KPI	Result
Revenue	208M
Profit	49M
Total Orders	8,000
Total Customers	500
Units Sold	17K
Returned Orders	651
Return Rate	8.14%

11. Business Insights

Electronics is the strongest category

Electronics generates the highest revenue and highest profit.

Laptop Pro 14 is the top-performing product

Laptop Pro 14 ranks first in both revenue and profit.

June 2024 recorded the highest revenue

June 2024 was the strongest revenue month.

Riya Das is the highest-value customer

Riya Das generated the highest customer revenue.

Overall return rate is 8.14%

651 of 8,000 orders were returned.

Wrong Product is the leading return reason

Wrong Product was the most common return reason.

Electronics has the most returned orders

Electronics combines the strongest sales performance with the highest return volume.

November 2025 had the highest return activity

November 2025 recorded the highest number of returns.

12. Business Recommendations

- Prioritize Electronicsinventory andavailability becauseit leads revenue and profit.
- Maintain healthy inventory for Laptop Pro 14 and evaluate targeted promotions.
- Investigate the drivers behind the June 2024 revenue peak and replicate successful strategies.
- Use loyalty or personalized retention initiatives for high-value customers.
- Investigate Wrong Product returns by reviewing product descriptions, SKU mapping, order picking and fulfillment.
- Compare Electronics return rate, not only return volume, to identify unusually high return performance.
- Investigate the November 2025 return spike by product, reason, promotion and fulfillment process.
- Continueusing Yearand Category slicers to compare business performance across periods and segments.

13. Conclusion

This project demonstrates a complete data analytics workflow from raw e-commerce transactions to business-ready insights. Python and Pandas supported data preparation, MySQL supported structured analysis, and Power BI transformed the results into an interactive dashboard.

The analysis identified Electronics and Laptop Pro 14 as major performance drivers, highlighted high-value customer behavior, and revealed return-related opportunities. The findings can support decisions around

inventory, salesstrategy, customer retentionand fulfillmentquality.

14. Project Completion Summary

Component	Status
Python / Pandas Data Cleaning	Completed
Feature Engineering	Completed
MySQL Business Analysis	Completed
Excel Data Preparation	Completed
Power BI Dashboard	Completed — 3 Pages
Dynamic DAX Measures	Completed
Business Insights	Completed
Recommendations	Completed